

ØNEIRO

Haptic Wristband Build Guide

XIAO ESP32C3 — Dual Motor — v2.0

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Overview

This guide covers the complete assembly of the Øneiro haptic wristband — a WiFi-triggered dual vibration motor system that receives REM detection signals from Phase Sync and fires haptic cues to induce lucidity during sleep. The wristband uses a Seeed Studio XIAO ESP32C3 microcontroller with two independent motor circuits controlled via GPIO D0 and D10.

Firmware is pre-flashed. Your job is to wire the circuit correctly, test each connection, and assemble the final wristband. Follow every step in order. Test continuity after every solder joint.

Parts List

Component	Qty	Notes
Seeed Studio XIAO ESP32C3 (pre-soldered headers)	1	Main microcontroller
Coin vibration motor with pre-soldered leads	2	One per wrist position — get motors with 50mm wire leads
2N2222 NPN transistor	2	One per motor circuit
1N4001 flyback diode	2	One per motor — CRITICAL safety component
220 ohm resistor	2	Base resistor for each transistor
TP4056 USB charging module	1	Battery charge management
3.7V LiPo battery (2000mAh)	1	Check polarity before connecting
Cloth wristband	1	Final mounting
Heat shrink tubing	Plenty	Cover every solder joint
Jumper wire / short wire lengths	~12	Solid core 24-28 AWG preferred

■ Critical Safety Rules — Read Before Touching Anything

- NEVER power up the circuit without first testing continuity between VUSB and GND — a short will instantly damage or destroy the XIAO.
- NEVER connect the LiPo battery without verifying polarity with a multimeter first. Reversed polarity destroys the TP4056 and battery immediately.
- If the 1N4001 diode gets warm on power-up — cut power instantly. The diode is reversed. Fix before continuing.
- The XIAO ESP32C3 does NOT support 5GHz WiFi. The router must broadcast on 2.4GHz.
- Test continuity after every single solder joint before moving to the next step.

2N2222 Transistor Pin Reference

Hold the transistor with the FLAT FACE toward you, pins pointing DOWN. Left to right:

Pin	Name	Connects To
Left pin (Pin 1)	Emitter	GND
Middle pin (Pin 2)	Base	220 ohm resistor from GPIO pin
Right pin (Pin 3)	Collector	Motor black wire (–)

1N4001 Diode Orientation — CRITICAL

The stripe on the 1N4001 marks the CATHODE. The diode sits in PARALLEL across the motor terminals — it is a protection component only, NOT in the main power path.

End	Name	Connects To
End WITH stripe	Cathode (+)	Motor red wire (+)
End WITHOUT stripe	Anode (–)	Motor black wire (–)

The diode connects ONLY to the two motor wires. Both legs bridge across the motor — cathode to red (+), anode to black (–). Nothing else.

Full Wiring Reference — Both Circuits

FROM	TO	Notes
XIAO GPIO D0	220 ohm resistor leg 1	Motor A control signal
220 ohm resistor leg 2	2N2222 Base (middle pin)	Current-limited gate — Motor A
2N2222 Emitter (left pin)	XIAO GND	Transistor return — Motor A
2N2222 Collector (right pin)	Motor A black wire (–)	Switched ground — Motor A
Motor A red wire (+)	XIAO VUSB	Motor A power
1N4001 cathode (stripe)	Motor A red wire (+)	Flyback protection — parallel with motor
1N4001 anode (no stripe)	Motor A black wire (–)	Flyback protection — parallel with motor
XIAO GPIO D10	220 ohm resistor leg 1	Motor B control signal
220 ohm resistor leg 2	2N2222 Base (middle pin)	Current-limited gate — Motor B
2N2222 Emitter (left pin)	XIAO GND	Transistor return — Motor B
2N2222 Collector (right pin)	Motor B black wire (–)	Switched ground — Motor B
Motor B red wire (+)	XIAO VUSB	Motor B power
1N4001 cathode (stripe)	Motor B red wire (+)	Flyback protection — parallel with motor
1N4001 anode (no stripe)	Motor B black wire (–)	Flyback protection — parallel with motor

LiPo battery (+)	TP4056 B+	Check polarity first
LiPo battery (-)	TP4056 B-	Check polarity first
TP4056 OUT+	XIAO VUSB pin	Main power in
TP4056 OUT-	XIAO GND	Common ground

Step-by-Step Build — Motor A First

Build Motor A completely and test it before starting Motor B. Test continuity after EVERY solder joint with multimeter in Ω) mode.

Step 1 — D0 wire

Solder a short wire to the XIAO D0 pin. D0 is on the LEFT side of the board, bottom-most pin above the USB-C port. Continuity test: probe D0 pin and free end of wire — should beep.

Step 2 — 220 ohm resistor

Solder one leg of the 220 ohm resistor to the free end of the D0 wire. Leave the other resistor leg free. Continuity: D0 to resistor leg 1 — beep.

Step 3 — Transistor Base

Solder the free resistor leg to the MIDDLE pin (Base) of the 2N2222. Hold transistor flat face toward you, pins down. Middle = Base. Continuity: D0 to Base — beep.

Step 4 — Transistor Emitter to GND

Solder a wire from the LEFT pin (Emitter) of the 2N2222 to the XIAO GND pin. Continuity: Emitter to GND — beep.

Step 5 — Motor black wire to Collector

Solder the motor BLACK wire (–) to the RIGHT pin (Collector) of the 2N2222. Continuity: Motor black wire to Collector — beep. If no beep — reflow the joint with fresh solder before continuing.

Step 6 — Motor red wire to VUSB

Solder the motor RED wire (+) to the XIAO VUSB pin. Continuity: Motor red wire to VUSB — beep.

Step 7 — Flyback diode

Solder the 1N4001 ACROSS the motor terminals. Cathode (stripe end) to motor RED wire. Anode (no stripe) to motor BLACK wire. The diode bridges the two motor wires — it is NOT in series with anything else.

Step 8 — Pre-power continuity check

BEFORE plugging in USB-C: Set multimeter to Ω) continuity mode. Probe VUSB and GND. Must be SILENT — no beep. If it beeps there is a short — do not power up.

Step 9 — Power up and test

Plug USB-C into XIAO. Open Arduino Serial Monitor at 115200 baud. Confirm board boots and connects to WiFi. Run: `cargo run --bin manual_trigger 3 500 255` from Phase Sync directory. Motor A should vibrate 3 times.

Motor B — Repeat Identically

Once Motor A is confirmed working, repeat Steps 1–9 identically for Motor B. The only difference is the control pin — use GPIO D10 instead of D0. D10 is on the LEFT side of the board, 4th pin from the top.

Firmware Configuration

Edit these constants at the top of oneiro_haptic_local.ino before flashing:

```
const char* WIFI_SSID = "YourNetworkName";  
const char* WIFI_PASSWORD = "YourPassword";  
const char* PC_IP = "192.168.x.x"; // IP of PC running Phase Sync
```

Constant	Value	Description
MOTOR_A_PIN	2	GPIO D0 — Motor A
MOTOR_B_PIN	10	GPIO D10 — Motor B
BATTERY_PIN	4	GPIO A2 — battery ADC
LEDC_CH_A	0	LEDC channel for Motor A
LEDC_CH_B	1	LEDC channel for Motor B
OSC_PORT	5001	Listens for Phase Sync triggers
ACK_PORT	5002	Sends ACK back to Phase Sync

Flash Instructions

1. Open Arduino IDE 2.x
2. Board: Tools → Board → ESP32 Arduino → XIAO_ESP32C3
3. Port: Tools → Port → COM7 (or whichever port appears after plugging in USB-C)
4. Click Upload
5. Open Serial Monitor at 115200 baud
6. Confirm: [Øneiro] Ready — waiting for Phase Sync trigger...

Troubleshooting

Symptom	Fix
Motor does not vibrate	Test continuity at every joint. Start at Collector-to-motor-black. Reflow any joint with no beep.
Motor runs constantly	Transistor wired backwards or base resistor missing. GPIO floating HIGH.
Diode gets hot on power-up	Diode reversed — cut power immediately. Stripe end must face motor (+).
Short between VUSB and GND	Check all solder joints for bridges. Do not power up until resolved.
Board boots but no WiFi	Must be 2.4GHz network. Check SSID and password in firmware.
Serial Monitor shows nothing	Wrong COM port. Try different USB-C cable (some are charge-only).
Motor vibrates weakly	Motor (+) must connect to VUSB not 3.3V. Check power connection.
No ACK in Phase Sync	Confirm PC_IP matches PC IPv4 address. Run ipconfig to verify.

OSC Protocol Reference

OSC Address	Args	Function
/oneiro/trigger	pulses, durationMs, strength	Fire haptic cue — both motors
/oneiro/timer/start	(none)	Start 60-second dream timer pulses
/oneiro/timer/stop	(none)	Stop dream timer
/oneiro/ack	pulses, durationMs, strength	ACK sent back to Phase Sync

Manual trigger command from Phase Sync directory:

```
cargo run --bin manual_trigger <pulses> <durationMs> <strength>
```

```
Example: cargo run --bin manual_trigger 3 500 255
```